

Foundation (Jalapeno)

- Q1.** In finance, if a stock's daily return is represented by x , what does $|x|$ represent?
- (A) The closing price of the stock
(B) The magnitude of the price movement regardless of direction
(C) The total profit earned today
(D) undefined
(E) The ratio of gains to losses
- Q2.** A bank account shows a transaction of $-\$450.25$. What is the absolute value of this transaction?
- (A) \$0
(B) $-\$450.25$
(C) \$450.25
(D) \$900.50
(E) \$1
- Q3.** Evaluate the expression: $|-12| + |7| - |-3|$.
- (A) 16
(B) 22
(C) 8
(D) 2
(E) 10
- Q4.** If a tax refund discrepancy is modeled as $|d| = 15$, which values could represent the discrepancy d ?
- (A) 15 only
(B) -15 only
(C) 15 and -15
(D) 0 and 15
(E) No solution
- Q5.** Which of the following represents the distance between a debt of $-\$20$ and a profit of $\$30$ on a number line?
- (A) $|30 - 20|$
(B) $|-20 + 30|$
(C) $|30 - (-20)|$
(D) $|-20| - |30|$
(E) $|20| + |-30|$
- Q6.** A hedge fund manager observes a price change p . If $|p| = 0$, what does this indicate about the asset price?
- (A) The price doubled
(B) The price fell to zero
(C) The price remained unchanged
(D) The price hit a record high
(E) The market is closed
- Q7.** Simplify the financial ratio expression: $\frac{|-100|}{|-25|}$.
- (A) -4
(B) 4
(C) 0.25
(D) -0.25
(E) 125
- Q8.** Compare the following magnitudes: $A = |-55|$ and $B = |42|$. Which statement is true?
- (A) $A < B$
(B) $A = B$
(C) $A > B$
(D) $A + B = 13$
(E) $A \cdot B < 0$

Open Ended Questions (FRQ)

- Q9.** A business budget has a planned expense of $\$5,000$. The actual expense was $\$4,850$. Calculate the absolute deviation using the formula $|Actual - Planned|$ and explain why absolute value is used in this context.
- Q10.** Evaluate the complex absolute value expression for a tax calculation: $|-150| - |25 - 40| + |-5|$. Show each step of your calculation.

Chef's Tips

- Emphasize that absolute value always results in a non-negative distance.
- Use the number line to visualize the distance from zero for negative stock returns.
- Remind students that $|a - b|$ is the distance between two points a and b .